Surface Mount Schottky Barrier Rectifier

SSA24

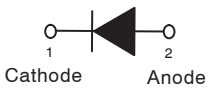


SMA
CASE 403AE

Features

- UL Flammability 94V−0 Classification
- MSL 1
- Industrial Device Qualified per AEC−Q101 Standards.
  * see authorized use policy
- This Device is Pb−Free, Halogen Free and are RoHS Compliant

CONNECTION DIAGRAM



MARKING DIAGRAM



(Color Band Denotes Cathode)

&Z = Assembly Plant Code
&3 = Numeric Date Code (Year & Week)
SSA24 = Specific Device Code

ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Value	Unit
V_{RRM}	Recurrent Peak Reverse Voltage	40	V
V_{RMS}	RMS Voltage	28	V
V_{DC}	DC Blocking Voltage	40	V
$I_{F(AV)}$	Average Forward Current at T_L = 75°C	2	A
I_{FSM}	Peak Forward Surge Current: 8.3 ms Single Half Sine−Wave Superimposed on Rated Load (JEDEC Method)	50	A
T_J	Operating Junction Temperature Range	−55 to +150	°C
T_{STG}	Storage Temperature Range	−55 to +150	°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Value	Unit
Ψ_{JL}	Typical Thermal Characteristics, Junction−to−Lead (Note 1)	20	°C/W
$R_{\theta JA}$	Typical Thermal Resistance, Junction−to−Ambient (Note 1)	75	°C/W

1. Mounted on P. C. Board with 8 mm^2 (0.013 mm thick) Copper pad Areas.

ELECTRICAL CHARACTERISTICS

(Values are at T_A = 25°C unless otherwise noted.)

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
V_F	Forward Voltage (Note 2)	I_F = 2.0 A	–	–	0.5	V
I_R	DC Reverse Current	V_R = 40 V	–	–	0.2	mA
		V_R = 40 V, T_A = 100°C	–	–	20	
T_{rr}	Reverse Recovery Time	I_F = 0.5 A, I_R = 1 A, I_{rr} = 0.25 A	–	9.84	–	nA

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. Pulse Test Pulse Width = 300 μs,1% Duty Cycle.

ORDERING INFORMATION

Device	Package	Shipping†
SSA24	SMA (Pb−Free)	7500 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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TYPICAL PERFORMANCE CHARACTERISTICS

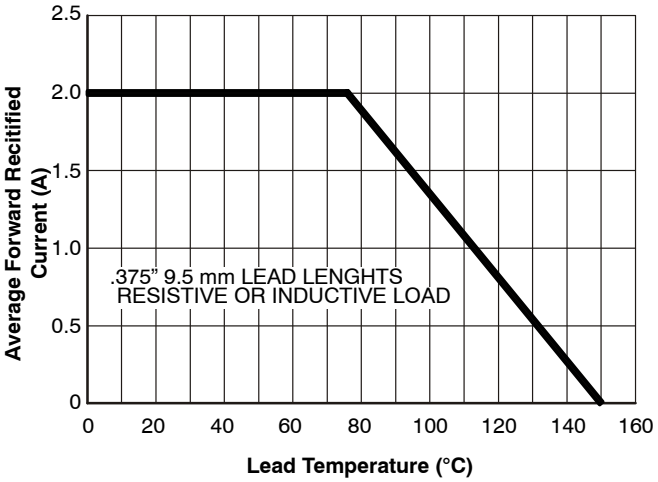


Figure 1. Forward Current Derating Curve

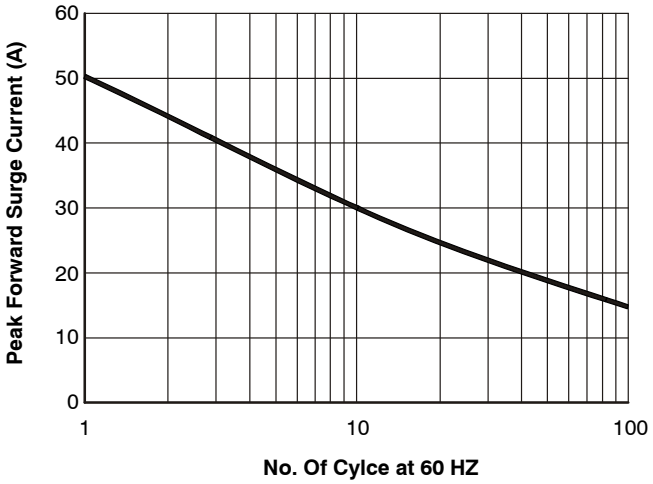


Figure 2. Maximum Non−Repetitive Peak Forward Surge Current

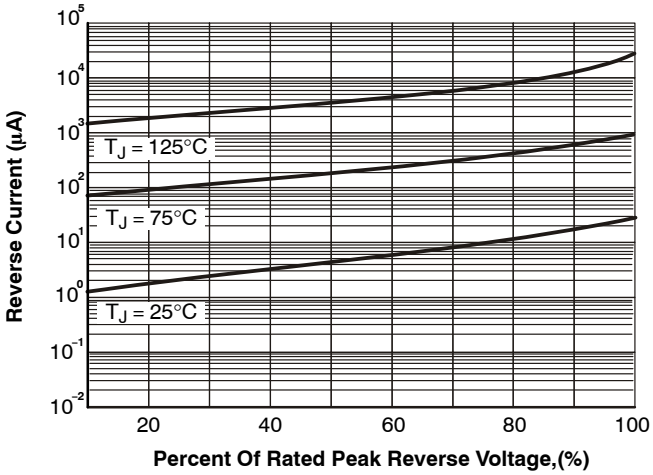


Figure 3. Typical Reverse Characteristic

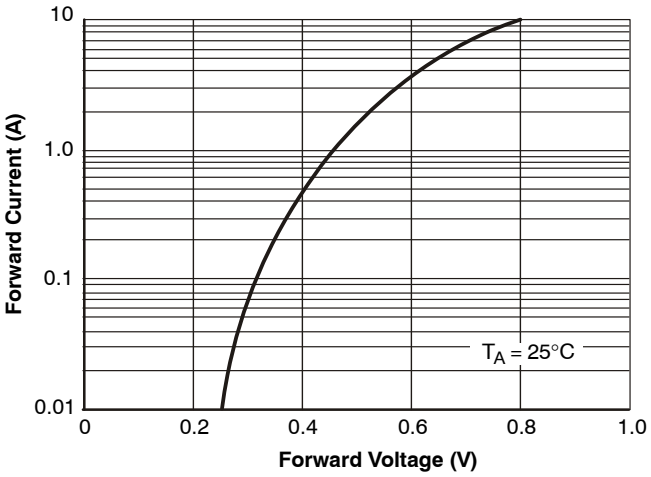


Figure 4. Typical Instantaneous Forward Characteristics

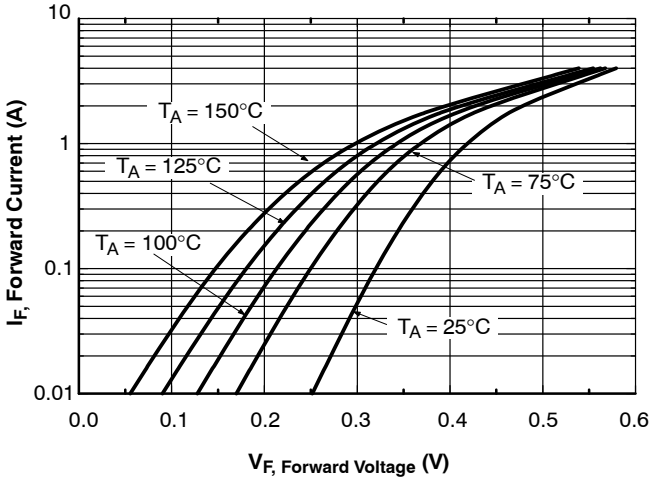


Figure 5. Typical Forward Characteristics

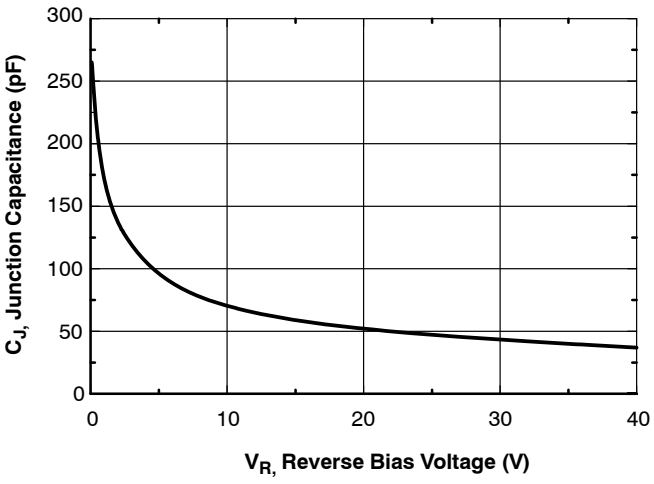
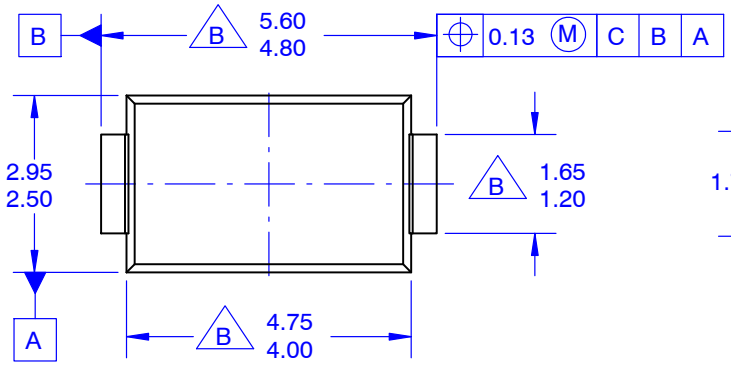


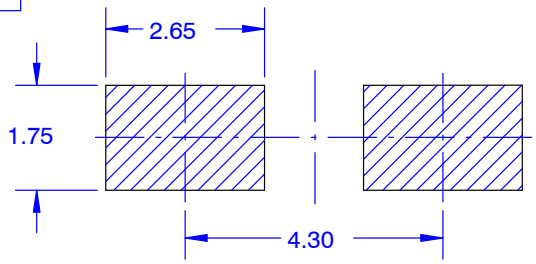
Figure 6. Typical Junction Capacitance

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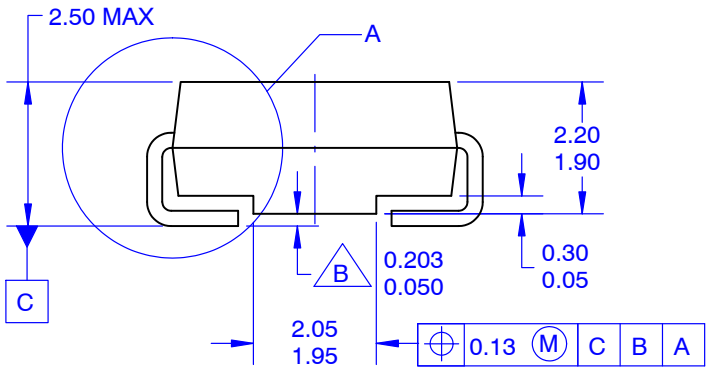
DATE 31 AUG 2016



TOP VIEW



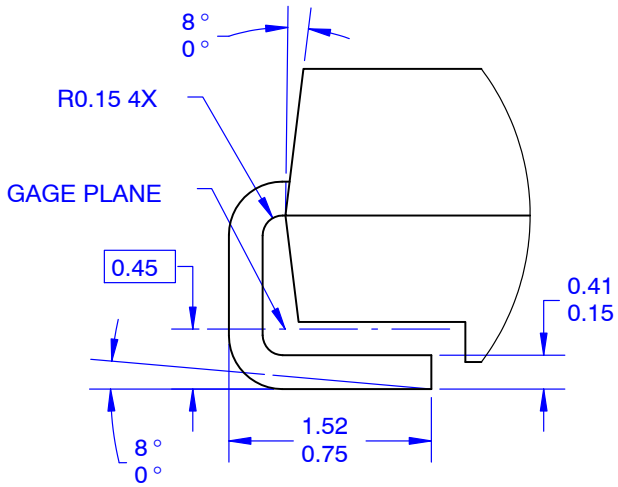
LAND PATTERN RECOMMENDATION



SIDE VIEW

NOTES:

A. EXCEPT WHERE NOTED, CONFORMS TO JEDEC DO214 VARIATION AC.
B DOES NOT COMPLY JEDEC STANDARD VALUE.
C. ALL DIMENSIONS ARE IN MILLIMETERS.
D. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH AND TIE BAR PROTRUSIONS.
E. DIMENSIONS AND TOLERANCE AS PER ASME Y14.5–2009.
E. LAND PATTERN STD. DIOM5025X231M



DETAIL A
SCALE 20 : 1

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